Project Initialization and Planning Phase

Template: Project Proposal (Proposed Solution) Report	Maximum Marks: 5 Marks
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Project Proposal (Proposed Solution) Report

The proposal report aims to transform food intelligence and culinary health using multimodal Generative AI and computer vision, boosting operational efficiency, nutritional precision, and waste reduction. It tackles system inefficiencies, promising better lifestyle health, reduced food waste, and verified hygiene compliance. Key features include a multimodal food recognition model, OpenCV decay feature extraction, an interactive hands-free cooking assistant, and an automated clinical meal planner.

Project Overview & Problem Statement:

Objective: Revolutionize nutritional tracking, produce shelf-life inspection, commercial kitchen auditing, and clinical meal planning through a single integrated multimodal agent platform.
Scope: Comprehensive full-stack implementation comprising FastAPI backend microservices, React 18 / Vite modern frontend, OpenCV image processing, and cloud LLM reasoning deployed live on Vercel and Render.
Impact: Delivers automated food macro breakdown in under 3 seconds, eliminates household produce spoilage through biochemical decay warnings, guarantees zero-allergen cross-contamination in meal planning, and digitizes restaurant safety compliance.

Proposed Solution Approach & Key Features:

• **Multimodal Vision-Language Reasoning:** Employs Groq Qwen 3.8B Multimodal model to identify complex food plates and estimate calories, protein, carbs, fats, fiber, and allergens from image and audio voice notes.
• **Computer Vision Spoilage Quantification:** Utilizes OpenCV HSV color uniformity scoring and Laplacian gradient variance to measure produce browning and surface wrinkling objectively.
• **Real-Time Kitchen Hygiene Auditing:** Extracts representative video frames from commercial kitchen footage to audit staff PPE compliance (hairnets, gloves) and workstation cleanliness.
• **Conversational Culinary Assistant:** Real-time chef Q&A; engine that answers mid-cooking questions hands-free and generates adaptive zero-waste recipes.
• **Scientific Metabolic Planner:** Implements Mifflin-St Jeor metabolic equations to calibrate personalized BMR/TDEE meal schedules with strict multi-allergy exclusions.

Resource Requirements:

Resource Type	Description	Specification / Allocation (FoodVerse AI)
Hardware & Compute	Cloud Server CPU / Memory	Render Cloud Standard Tier (512MB RAM, shared vCPU) + Client Device Camera & Audio Mic
Software & Frameworks	Backend & Frontend Frameworks	Python 3.10+, FastAPI 0.149, Uvicorn ASGI Server, React 18.3, Vite 6.0
AI & CV Libraries	Deep Learning & Vision Engines	Groq SDK 0.11+, OpenCV headless 4.10+, NumPy, Pillow, SpeechRecognition, PyDub
Development & Cloud	IDE, Version Control, Hosting	VS Code, Git/GitHub, Vercel Edge CDN (Frontend), Render Web Services (Backend)
Data & Model Sources	Inference Models & Benchmarks	Groq Qwen 3.8B Multimodal Vision-Language Model, Whisper Large v3 Turbo, Mifflin-St Jeor Dietetics